

Decent Coaching Classes

VIII CBSE

Mind Your Math - V

ANSWER KEY

Marks: 30

Time: 1 hr.

Section A – Multiple Choice Questions (1 mark each)

[6]

1. Answer: (B) $49y^2 + 84y + 36$. $[(a + b)^2 = a^2 + 2ab + b^2 = (7y)^2 + 2(7y)(6) + 6^2]$
2. Answer: (C) 9996. $[98 \times 102 = (100 - 2)(100 + 2) = 100^2 - 2^2 = 10000 - 4 = 9996]$
3. Answer: (A) $20x^3 + 12x^2$. $[4x \times 5x^2 = 20x^3; 4x \times 3x = 12x^2]$
4. Answer: (D) $y^4 - x^4$. $[(a + b)(a - b) = a^2 - b^2, \text{ with } a = y^2 \text{ and } b = x^2]$
5. Answer: (B) 9200. $[96^2 - 4^2 = (96 - 4)(96 + 4) = 92 \times 100 = 9200]$
6. Answer: (C) 2728. $[248 \times 11 = 248 \times (10 + 1) = 2480 + 248 = 2728]$

Section B – Very Short Answer Questions (2 marks each)

[10]

7. $(5/7)(14x - 7y + 3/5)$
 $= (5/7)(14x) + (5/7)(-7y) + (5/7)(3/5)$ (1 mark)
 $= 10x - 5y + 3/7$ (1 mark)
8. Product in one line
(i) $6314 \times 11 = 6314 \times (10 + 1) = 63140 + 6314 = 69454$ (1 mark)
(ii) $2865 \times 101 = 2865 \times (100 + 1) = 286500 + 2865 = 289365$ (1 mark)
9. 6725×99
 $6725 \times 99 = 6725 \times (100 - 1)$ (1 mark)
 $= 672500 - 6725$
 $= 665775$ (1 mark)
10. $(3x^2y - 2y^2)^2$
Using $(a - b)^2 = a^2 - 2ab + b^2$, with $a = 3x^2y$ and $b = 2y^2$ (1 mark)
 $= (3x^2y)^2 - 2(3x^2y)(2y^2) + (2y^2)^2$
 $= 9x^4y^2 - 12x^2y^3 + 4y^4$ (1 mark)
11. $23x = 68^2 - 45^2$
 $68^2 - 45^2 = (68 - 45)(68 + 45) = 23 \times 113$ (1 mark)
So $23x = 23 \times 113$
 $x = 113$ (1 mark)

Section C – Short Answer Questions (4 marks each)

[8]

12. Simplify : $(x + 2y)(3 + 2x - y) - 2x(x + y)$
 $(x + 2y)(3 + 2x - y) = x(3 + 2x - y) + 2y(3 + 2x - y)$
 $= 3x + 2x^2 - xy + 6y + 4xy - 2y^2$
 $= 2x^2 + 3xy - 2y^2 + 3x + 6y$ (2 marks)
 $2x(x + y) = 2x^2 + 2xy$
Required expression = $(2x^2 + 3xy - 2y^2 + 3x + 6y) - (2x^2 + 2xy)$ (1 mark)
 $= xy - 2y^2 + 3x + 6y$ (1 mark)

OR

- Simplify : $(3a - 2b)^2 - (3a + 2b)^2$
 $(3a - 2b)^2 = 9a^2 - 12ab + 4b^2$ (1 mark)
 $(3a + 2b)^2 = 9a^2 + 12ab + 4b^2$ (1 mark)

$$\text{Difference} = (9a^2 - 12ab + 4b^2) - (9a^2 + 12ab + 4b^2) \quad (1 \text{ mark})$$

$$= -24ab \quad (1 \text{ mark})$$

$$13. (3x - 2)(3x + 2)(9x^2 + 4)$$

$$(3x - 2)(3x + 2) = (3x)^2 - 2^2 = 9x^2 - 4 \quad [\text{using } (a + b)(a - b) = a^2 - b^2] \quad (1 \text{ mark})$$

$$(9x^2 - 4)(9x^2 + 4) = (9x^2)^2 - 4^2 \quad (1 \text{ mark})$$

$$= 81x^4 - 16$$

Verification for $x = 1$:

$$\text{LHS} = (3 - 2)(3 + 2)(9 + 4) = 1 \times 5 \times 13 = 65$$

$$\text{RHS} = 81(1)^4 - 16 = 65 \quad (1 \text{ mark})$$

LHS = RHS, hence verified. (1 mark)

OR

$$3x^5 \times (-1/9)xy^2 \times 4xyz^2, \text{ when } x = 1, y = 2, z = 3$$

$$\text{Coefficients : } 3 \times (-1/9) \times 4 = -4/3 \quad (1 \text{ mark})$$

$$\text{Variables : } x^5 \times x \times x = x^7; \quad y^2 \times y = y^3; \quad z^2 \quad (1 \text{ mark})$$

$$\text{Product} = -(4/3)x^7y^3z^2 \quad (1 \text{ mark})$$

$$\text{Put } x = 1, y = 2, z = 3 : -(4/3)(1)^7(2)^3(3)^2 = -(4/3) \times 8 \times 9 = -(4/3) \times 72$$

$$= -96 \quad (1 \text{ mark})$$

Section D – Case-Based Question (6 marks)

[6]

14. Garden and play area

(i) The identity $(a + b)(a - b) = a^2 - b^2$ (identity 1C : difference of squares), because $98 = 100 - 2$ and $102 = 100 + 2$. (1 mark)

$$(ii) \text{ Area of garden} = 98 \times 102 = (100 - 2)(100 + 2) \quad (1 \text{ mark})$$

$$= 100^2 - 2^2 = 10000 - 4 = 9996 \text{ m}^2 \quad (1 \text{ mark})$$

$$(iii) \text{ Area of square play area} = 102^2 = (100 + 2)^2 \quad [\text{using } (a + b)^2 = a^2 + 2ab + b^2]$$

$$= 100^2 + 2 \times 100 \times 2 + 2^2 = 10000 + 400 + 4 = 10404 \text{ m}^2 \quad (2 \text{ marks})$$

Difference = $10404 - 9996 = 408 \text{ m}^2$. The play area is 408 m^2 larger than the garden. (1 mark)